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## **ME-7005(1)-CBGS**

### **B.E. VII Semester**

Examination, December 2020

## **Choice Based Grading System (CBGS)**

### **Robotics**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1.
  - a) Give all possible classification of robots.
  - b) Describe functions of four basic components of a robot.
2.
  - a) What is Repeatability? Discuss about accuracy and resolution.
  - b) Discuss various types of programming method used in robot.
3.
  - a) Explain important design considerations for robot grippers.
  - b) Define term end effectors. What factors should be embodied in the design of end effectors for use on industrial robots?
4.
  - a) Write down principle and applications of LVDT sensors.
  - b) Write down the difference between range sensors and proximity sensors.

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5. Discuss following sensors and their applications in Robotics.
  - a) Position sensors
  - b) Touch sensors
  - c) Slip sensors
6. Discuss concept of forward kinematics and reverse kinematics of manipulators with two and three degree of freedom.
7.
  - a) What problem will be encountered if the frames are arbitrarily assigned to develop the forward kinematic model of manipulator.
  - b) Why is it important to choose a frame assignment for an n-DOF manipulator that gives a maximum number of zero joint link parameters?
8. Write short technical note on following (Any two):
  - i) Work cycle time analysis for robot
  - ii) Economic considerations for robot
  - iii) Robotic softwares

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